

VULNERABILITY ASSESSMENT

# & Penetration Testing Report

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|             |                                    |
|-------------|------------------------------------|
| Scope       | Internal Network — 192.168.1.0/24  |
| Duration    | 5 Days                             |
| Methodology | PTES / OWASP                       |
| Tools Used  | Nmap, Wireshark, Metasploit, Nikto |

EXECUTIVE SUMMARY

Executive Summary

This report presents the findings of a vulnerability assessment and penetration test conducted on the internal network environment. The assessment identified multiple security weaknesses ranging from critical misconfigurations to low-risk informational disclosures. Immediate remediation is recommended for critical and high findings.

RISK SUMMARY

| Severity | Count | Status                    |
|----------|-------|---------------------------|
| CRITICAL | 2     | Requires immediate action |
| HIGH     | 4     | Remediate within 7 days   |
| MEDIUM   | 6     | Remediate within 30 days  |
| LOW      | 5     | Monitor and review        |
| INFO     | 8     | No action required        |

METHODOLOGY

Testing Methodology

1. Reconnaissance

Passive and active information gathering using Nmap and OSINT techniques to map the attack surface.

2. Scanning & Enumeration

Service discovery, OS fingerprinting, and open port analysis across all in-scope hosts.

3. Vulnerability Analysis

Automated scanning with Nikto and OpenVAS, cross-referenced with CVE database entries.

4. Exploitation

Controlled exploitation of confirmed vulnerabilities using Metasploit Framework.

5. Post-Exploitation

Privilege escalation attempts and lateral movement analysis within the test environment.

6. Reporting

Documentation of all findings with risk ratings, evidence, and remediation guidance.

FINDINGS

Detailed Findings

|                                                         |                                                                                        |
|---------------------------------------------------------|----------------------------------------------------------------------------------------|
| [CRITICAL] CVF-001 — Unpatched SMB Service (MS17-010)   |                                                                                        |
| Host / Asset                                            | 192.168.1.45                                                                           |
| Description                                             | Port 445 running SMBv1 with EternalBlue vulnerability present.                         |
| Remediation                                             | Disable SMBv1 immediately. Apply MS17-010 patch. Block port 445 at perimeter firewall. |
| [CRITICAL] CVF-002 — Default Credentials on Admin Panel |                                                                                        |
| Host / Asset                                            | 192.168.1.10                                                                           |
| Description                                             | Web admin interface accessible with default admin:admin credentials.                   |
| Remediation                                             | Change all default credentials. Implement account lockout policy. Enable MFA.          |
| [HIGH] CVF-003 — SSL/TLS Certificate Expired            |                                                                                        |
| Host / Asset                                            | 192.168.1.22                                                                           |
| Description                                             | HTTPS service running with expired certificate — MITM risk.                            |
| Remediation                                             | Renew SSL certificate. Implement automated renewal with Let's Encrypt.                 |
| [HIGH] CVF-004 — Open Telnet Service                    |                                                                                        |
| Host / Asset                                            | 192.168.1.33                                                                           |
| Description                                             | Telnet (port 23) transmits credentials in plaintext.                                   |
| Remediation                                             | Disable Telnet. Replace with SSH. Update firewall rules to block port 23.              |

RECOMMENDATIONS

Remediation Roadmap

| Priority | Action                             | Timeline  | Owner         |
|----------|------------------------------------|-----------|---------------|
| Critical | Patch MS17-010 / Disable SMBv1     | Immediate | Sys Admin     |
| Critical | Change all default credentials     | Immediate | Sec Team      |
| High     | Renew SSL certificates             | 7 Days    | DevOps        |
| High     | Disable Telnet, enable SSH         | 7 Days    | Sys Admin     |
| Medium   | Implement network segmentation     | 30 Days   | Network Team  |
| Low      | Update security awareness training | 90 Days   | HR / Sec Team |